

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 1

2024

1. Warm-up question

- (a) Show that $|a + b| \leq |a| + |b|$ for all $a, b \in \mathbb{R}$.
- (b) For $x, y \in \mathbb{R}$, define $p(x, y) = |\arctan x - \arctan y|$. Prove that p is a metric on $\mathbb{R}$.

2. For $\mathbf{a} = (a_1, a_2)$ and $\mathbf{b} = (b_1, b_2)$ in $\mathbf{R}^2$, put

$$\begin{aligned}d_m(\mathbf{a}, \mathbf{b}) &= \max\{|a_1 - b_1|, |a_2 - b_2|\} \\d_E(\mathbf{a}, \mathbf{b}) &= \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2} \\d_1(\mathbf{a}, \mathbf{b}) &= |a_1 - b_1| + |a_2 - b_2|\end{aligned}$$

- (a) Both d_E and d_1 are metrics on $\mathbf{R}^2$ (see Examples 2.4 and 2.7). Let's take that as given for now. Is d_m a metric on $\mathbf{R}^2$? Prove your answer.
 - (b) Let $\mathbf{0} = (0, 0)$. Sketch the set $\{\mathbf{x} \in \mathbf{R}^2 : d_m(\mathbf{0}, \mathbf{x}) \leq 1\}$.
 - (c) Sketch the sets obtained by replacing d_m in (b) by (i) d_E and (ii) d_1 .
3. Definition: A function $d : X \times X \rightarrow \mathbf{R}$ which satisfies (M1), the separation property, (M2), the symmetry property and the additional condition that

$$(\text{UM3}) \quad d(x, z) \leq \max \{d(x, y), d(y, z)\} \text{ for all } x, y, z \in X$$

is called an **ultrametric** on the set X . (We'll find ultrametrics very useful as examples and counter-examples.)

- (a) Check quickly that if d satisfies (UM3), then it automatically satisfies the (M3), the triangle inequality. (You can probably do this in your head.)

- (b) **Warm-up question** Prove that the Euclidean metric on $\mathbf{R}$ is not an ultrametric. Here $d_E(x, y) = |x - y|$, of course.

Let X be a non-empty set and $\mathcal{S}_X$ the collection of all sequences of elements of X . Define a metric d on $\mathcal{S}_X$ as follows:

$$d((x_n), (y_n)) = \begin{cases} \frac{1}{\min\{n \mid x_n \neq y_n\}} & \text{if } (x_n) \neq (y_n) \\ 0 & \text{if } (x_n) = (y_n) \end{cases}$$

- (c) To get an idea of what's going on here, let $X = \{1, 2, 3, 4, 5\}$. Make up a few elements of $\mathcal{S}_X$ for this X , and find the distances between them, as given by this d .

A remark about what a sequence is: The definition of a sequence of elements of X is a function from the natural numbers to X . We could informally think of it as an infinite ordered list $(x_n) = (x_1, x_2, x_3, \dots)$ where each x_n is an element of X . We should emphasize that there is an x_n for *every* natural number n ; in other words, we always use the word “sequence” to mean “infinite sequence” and of course, this doesn't mean X has to be infinite, the same element of X can appear in a sequence many times.

- (d) Returning to the general case now (meaning take X arbitrary and not $X = \{1, 2, 3, 4, 5\}$ anymore): prove that d is an ultrametric on $\mathcal{S}_X$.

4. Definition: A **pseudometric** on a set X is a function $d : X \times X \rightarrow \{x \in \mathbf{R} \mid x \geq 0\}$ such that for all x, y and z in X :

(PM1) $x = y$ implies that $d(x, y) = 0$

(PM2) $d(y, x) = d(x, y)$

(PM3) $d(x, z) \leq d(x, y) + d(y, z)$

The only difference between metric and pseudometric occurs in the conditions (M1) and (PM1).

Suppose (X, d) is a pseudometric space. Define a relation $\sim$ on X as follows. For all x and y in X ,

$$x \sim y \text{ iff } d(x, y) = 0.$$

- (a) Prove that $\sim$ is an equivalence relation on X (i.e. is reflexive, symmetric and transitive).
- (b) Denote the equivalence class containing x with respect to the relation $\sim$ by $[x]$. Define δ as follows. For $x, y \in X$,

$$\delta([x], [y]) = d(x, y).$$

- i. Show that δ is well-defined; in other words, if $[x] = [a]$ and $[y] = [b]$, then $\delta([x], [y]) = \delta([a], [b])$. (You are checking that the distance between equivalence classes doesn't depend on the representative of the class you choose.)
- ii. Prove that δ is a metric on the set of equivalence classes of X .

What you've done here is to change a pseudometric into a metric by "identifying the points of distance zero apart."

5. Here is an outline of how one might prove the triangle inequality for the Euclidean metric d_E in $\mathbf{R}^n$ (see Example 2.6 in the notes):

(a) We want to show

$$d_E(\mathbf{x}, \mathbf{z}) \leq d_E(\mathbf{x}, \mathbf{y}) + d_E(\mathbf{y}, \mathbf{z}).$$

Set $r_i = x_i - y_i$ and $s_i = y_i - z_i$. Then this is equivalent to showing:

$$\left(\sum (r_i + s_i)^2\right)^{\frac{1}{2}} \leq \left(\sum r_i^2\right)^{\frac{1}{2}} + \left(\sum s_i^2\right)^{\frac{1}{2}}$$

(b) Show this is equivalent to the Cauchy-Schwarz inequality:

$$\left(\sum (r_i s_i)\right)^2 \leq \left(\sum r_i^2\right) \left(\sum s_i^2\right)$$

(c) Now what could be the tricky bit: prove the Cauchy-Schwarz inequality. Hint: Consider the quadratic polynomial in z given by

$$p(z) = \sum_{1 \leq i \leq n} (r_i z + s_i)^2$$

and analyze its discriminant. (This may not be the most obvious thing to try, but it works easily.)

6. **Optional Challenge Question** Let X be the interval $(-1, 1)$ in the real line.

We define a new metric δ on X as follows:

$$\delta(P, Q) = \left| \ln \frac{(1-Q)(1+P)}{(1+Q)(1-P)} \right| \text{ for all } P, Q \in X.$$

- (a) Calculate $\delta(0, 0.1)$, $\delta(0, 0.9)$ and $\delta(0, 0.999)$.
- (b) Check that δ is a metric on X .
- (c) If $P < R < Q$, is it true that $\delta(P, R) + \delta(R, Q) = \delta(P, Q)$?
- (d) An ant is walking along the real line. It starts at 0 and heads towards the point 1. With every step it covers a distance of 1 (measured with respect to δ); and it takes a step a second. If it gets so close to the point 1 that it cannot take another step of distance 1, it stops. Has our ant stopped after 3 hours? [Hint: Before you do this, you need to be convinced that the correct answer to (c) above was "yes."]
- (e) Suppose (X, d) is a metric space and Y is a subset of X . If the supremum $D = \sup\{d(y_1, y_2) \mid y_1, y_2 \in Y\}$ exists, we say that the set Y has **diameter** D . What is the diameter of $(-1, 1)$ when the metric in use is (i) the Euclidean metric d_E (ii) the metric δ defined above?